

Clemens Thalhammer

PHD STUDENT · APPLIED MATHEMATICS

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Education

ETH Zurich

PHD APPLIED MATHEMATICS

- Supervisor: Habib Ammari

Zurich, Switzerland

since 2025

ETH Zurich

MSc APPLIED MATHEMATICS

- Focus in Mathematical Physics, Graduated with Distinction. Grade 5.78/6

Zurich, Switzerland

2023 - 2025

ETH Zurich

BSc MATHEMATICS

- Grade 5.24/6

Zurich, Switzerland

2020 - 2023

Professional Experience

2022, 2023 **Undergraduate Teaching Assistant**, ETH Zurich

2017-2019 **Coordinator**, Kinderuni Medizin, Medical University of Vienna

2013-2017 **Volunteer**, Kinderuni Medizin, Medical University of Vienna

Research Experience

Imperial College London - Dept. of Mathematics

ADVISORS: PROF. HABIB AMMARI, PROF. RICHARD CRASTER, DR. BRYN DAVIES

- MSc Thesis: "Computational Models for Quasiperiodic Differential Equations"

London, United Kingdom

Summer 2024

Teaching Experience

2023 **Numerical Methods**, Exam Preparation Course

2023 **Numerical Methods**, Teaching Assistant

2022 **Numerical Methods**, Teaching Assistant

Awards, Fellowships, & Grants

2025 **ETH Medal for an outstanding Master Thesis**, ETH Zurich

Presentations

INVITED TALKS

Summer 2026. **A Real-Space Formulation of the Zak Phase via Weyl m-Functions**. Invited talk: MetaMAT, Ajaccio, France.

Spring 2025. **Computational Models for Quasiperiodic ODEs**. Invited talk: Workshop on Metamaterials with Extra Dimensions, London, United Kingdom.

Leadership & Service

Seminar of Applied Mathematics, ETH Zurich

VICE GROUP ORGANISER

Oversaw the hiring of teaching assistants and coordinated the planning and distribution of teaching duties among PhD students. Contributed to administrative processes related to course organisation and delivery, ensuring smooth coordination and execution.

Zurich, Switzerland

since 2026

VMM (Mid-level Faculty Association), ETH Zurich*Zurich, Switzerland***CO-PRESIDENT***since 2025*

Represented early-career researchers in faculty governance and contributed to policy development and strategic decisions. Acted as a liaison between faculty leadership and members, ensuring transparent communication and engagement.

Kinderuni Medizin, Medical University of Vienna*Vienna, Austria***COORDINATOR***2017-2019*

Led the planning and execution of workshops for children, coordinating volunteers, scheduling events, and ensuring smooth program delivery for large audiences.

Publications

Habib Ammari, Silvio Barandun, Yannick De Bruijn, Ping Liu, Clemens Thalhammer. 2025. Spectra and pseudo-spectra of tridiagonal k -Toeplitz matrices and the topological origin of the non-Hermitian skin effect. *Journal of Physics A: Mathematical and Theoretical* 58(20):205201.

Bryn Davies, Clemens Thalhammer. 2025. Convergence of supercell and superspace methods for computing spectra of quasiperiodic operators. *Proceedings of the Royal Society A* 481(2319):20240894.

Habib Ammari, Silvio Barandun, Clemens Thalhammer, Alexander Uhlmann. 2026. Competing Edge and Bulk Localization in Nonreciprocal Disordered Systems. *Multiscale Modeling & Simulation* 24(2):456–473.

Habib Ammari, Clemens Thalhammer. 2026. A real-space formulation of the Zak phase via Weyl m -functions. *Journal of Physics A: Mathematical and Theoretical*.

PREPRINTS

Habib Ammari, Clemens Thalhammer, Alexander Uhlmann. 2025. Uniform Hyperbolicity, Bandgaps and Edge Modes in Aperiodic Systems of Subwavelength Resonators. *arXiv:2509.22417*.